

Machine Learning Syllabus (25 Topics)

1. Introduction to Artificial Intelligence and Machine Learning
2. Machine Learning Workflow
3. Types of Machine Learning
4. Python Programming for Machine Learning
5. NumPy and Pandas
6. Data Collection Techniques
7. Data Cleaning and Preprocessing
8. Exploratory Data Analysis (EDA)
9. Data Visualization
10. Statistics for Machine Learning
11. Probability Theory
12. Linear Algebra Fundamentals
13. Feature Engineering
14. Feature Selection
15. Supervised Learning
16. Linear Regression
17. Logistic Regression
18. Decision Trees
19. Random Forest
20. Support Vector Machines (SVM)
21. K-Nearest Neighbors (KNN)
22. Naive Bayes
23. Unsupervised Learning
24. Clustering Algorithms (K-Means, Hierarchical)
25. Dimensionality Reduction (PCA)
26. Neural Networks Basics
27. Deep Learning Fundamentals
28. Model Evaluation Metrics
29. Hyperparameter Tuning
30. Machine Learning Project Deployment